

490-HW2

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DURATION

```
# a
library(meta)

## Warning: package 'meta' was built under R version 4.2.2

## Loading 'meta' package (version 6.1-0).
## Type 'help(meta)' for a brief overview.
## Readers of 'Meta-Analysis with R (Use R!)' should install
## older version of 'meta' package: https://tinyurl.com/dt4y5drs

e_mean1 = c(60,49,51,52,47,82)
e_sd1 = c(9,24,10,23,17,18)
e_n1 = c(17,45,33,454,43,12)
c_mean1 = c(76,57,47,68,58,73)
c_sd1 = c(10,20,16,34,18,15)
c_n1 = c(18,59,32,386,52,12)
meta1 = metacont(e_n1, e_mean1, e_sd1, c_n1, c_mean1, c_sd1, sm="SMD",
                 method.tau="DL", level=0.9, level.ma =0.9)
summary(meta1)

##           SMD           90%-CI %W(common) %W(random)
## 1 -1.6407 [-2.2934; -0.9880]      2.3      12.0
## 2 -0.3640 [-0.6923; -0.0357]      9.1      18.8
## 3  0.2973 [-0.1132;  0.7077]      5.8      16.9
## 4 -0.5592 [-0.6753; -0.4432]     72.6      22.6
## 5 -0.6215 [-0.9689; -0.2741]      8.1      18.3
## 6  0.5244 [-0.1606;  1.2095]      2.1      11.4
##
## Number of studies combined: k = 6
## Number of observations: o = 1163
##
##           SMD           90%-CI      z  p-value
## Common effect model -0.4991 [-0.5980; -0.4002] -8.30 < 0.0001
## Random effects model -0.3948 [-0.7194; -0.0702] -2.00  0.0454
##
## Quantifying heterogeneity:
```

```
## tau^2 = 0.1677 [0.0765; 2.2451]; tau = 0.4095 [0.2767; 1.4984]
## I^2 = 80.8% [63.4%; 89.9%]; H = 2.28 [1.65; 3.15]
##
## Test of heterogeneity:
##      Q d.f.  p-value
## 26.02    5 < 0.0001
##
## Details on meta-analytical method:
## - Inverse variance method
## - DerSimonian-Laird estimator for tau^2
## - Jackson method for confidence interval of tau^2 and tau
## - Hedges' g (bias corrected standardised mean difference; using exact formulae)
```

```
# Looking at the 90% confidence intervals around the SMD for each study
# indicates that all studies showed a significant difference between the control
# group and experimental group except for studies 3 and 6. These studies had
# confidence intervals that included 0, implying a high p-value for the null
# hypothesis mean_exp = mean_cont, in turn implying there is not a significant
# difference between the two groups. Also worth nothing is that study 4 had a
# significant higher sample size, increasing the precision of its results and
# thus augmenting its reliability along with weight in the meta-analysis.
cat('Individual study p-values:',meta1$pval)
```

```
## Individual study p-values: 3.550072e-05 0.06817373 0.2335742 2.289507e-15 0.003251896 0.2079439
```

```
# The above output shows the p-values for each study, and affirms the conclusion
# drawn from the confidence intervals in the metacont output. That is, studies
# 3 and 6 have the lowest p-values, both around 0.2, which are insignificant at
# alpha = 0.1
```

```
# b
DL_tau = metacont(e_n1, e_mean1, e_sd1, c_n1, c_mean1, c_sd1, sm="SMD",
                 method.tau="DL")$tau # ask for tau
PM_tau = metacont(e_n1, e_mean1, e_sd1, c_n1, c_mean1, c_sd1, sm="SMD",
                 method.tau="PM")$tau
ML_tau = metacont(e_n1, e_mean1, e_sd1, c_n1, c_mean1, c_sd1, sm="SMD",
                 method.tau="ML")$tau
lower_tau1 = c(0.2767,0.3124,0.3124) # low 95% estimates for tau, each method
upper_tau1 = c(1.4984,1.8475,1.8475)
tau_estimates1 = rbind(DL_tau,PM_tau,ML_tau) # combine tau estimates
summary1 = data.frame(tau_estimates1,lower_tau1,upper_tau1) # combine all
summary1
```

```
##      tau_estimates1 lower_tau1 upper_tau1
## DL_tau      0.4094755      0.2767      1.4984
## PM_tau      0.6717859      0.3124      1.8475
## ML_tau      0.5294535      0.3124      1.8475
```

```
# The estimates for tau^2, while different with all three methods, are not
# majorly different in that each have confidence intervals that do not include
# 0, suggesting the between-study variability is significantly different to 0.
# That is, all methods for estimating tau^2 suggest that there is significant
```

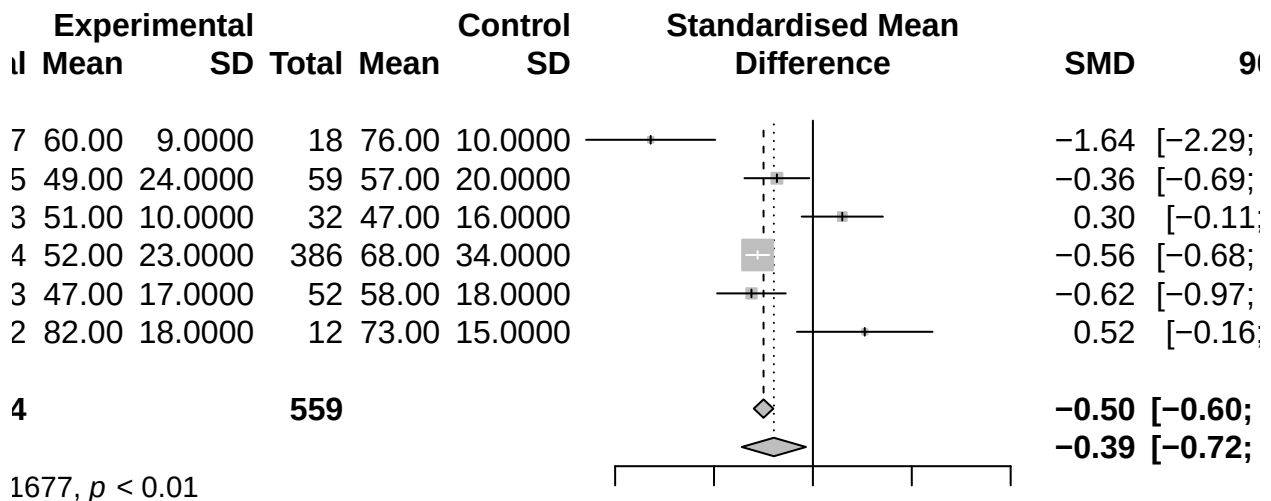
```
# variability between the studies.
# Extracting the confidence 90% confidence intervals from each method, it can be
# seen that they are relatively similar, albeit with a higher and wider interval
# for Paule-Mandel and restricted MLE methods. Also, PM and ML methods for tau^2
# had the exact same confidence interval, despite different point estimates.
# This could be due to higher variability in one of the methods.
```

```
# c
# For i = 1,2,...k studies, theta = observed standardized mean difference
# H_0: theta_1 = theta_2 = ... = theta_k vs. H_1: at least one inequality
Q1 = metacont(e_n1, e_mean1, e_sd1, c_n1, c_mean1, c_sd1, sm="SMD",
              method.tau="DL")$Q # get Q value from metacont
cat('Cochran p-value:', pchisq(Q1,5,lower.tail=FALSE)) # calculate p-value
```

```
## Cochran p-value: 8.835833e-05
```

```
# Since the p-value is (far) less than alpha = 0.01, we reject the null
# hypothesis and conclude there is significant between-study variability. This
# means the final model should be a random effects model, because there is
# significant evidence suggesting there is not a single true effect size value
# across all studies.
```

```
# d using logical operator "random" in metacont()
forest(metacont(e_n1, e_mean1, e_sd1, c_n1, c_mean1, c_sd1, sm="SMD",
               method.tau="DL",level=0.9,level.ma =0.9,random=1))
```



```
# e
# The overall conclusion based on the random effects model is that there is a
# significant difference between the effect tubeless PCNL and standard PCNL on
# kidney stones. Specifically, the effect size was found to be negative, meaning
# tubeless PCNL leads to better health. Statistically, this can be said because
# the 90% confidence interval on the expected effect size does not include 0.
# This means the p-value for the null hypothesis stating  $E(SMD) = 0$  is very low,
# leading to the conclusion that it is significant different to 0. Much of this
# result stems from the findings in study 4, which had the largest sample size
# and thus the largest weight in the model. Since this study found a significant
# improvement resulting from tubeless PCNL (i.e. the lowest p-value of all), it
# had the effect of "pulling" the expected effect size further below 0, despite
# contrary results from studies 3 and 6 (which had lower sample sizes and thus
# less "pull" on the overall results)
```

LOS

```
# a
e_mean2 = c(38,85,35,34,82,22,37)
e_sd2 = c(10,23,11,17,24,4,24)
e_n2 = c(17,45,33,454,43,101,12)
c_mean2 = c(67,129,44,56,106,54,38)
c_sd2 = c(22,54,22,62,35,5,24)
c_n2 = c(18,59,32,386,52,101,12)
meta2 = metacont(e_n2, e_mean2, e_sd2, c_n2, c_mean2, c_sd2, sm="SMD",
                 method.tau="DL", level=0.9, level.ma =0.9)
summary(meta2)
```

```
##           SMD           90%-CI %W(common) %W(random)
## 1 -1.6419 [-2.2947; -0.9891]         2.3        13.9
## 2 -1.0056 [-1.3515; -0.6597]         8.0        14.6
## 3 -0.5137 [-0.9289; -0.0986]         5.6        14.4
## 4 -0.5013 [-0.6170; -0.3857]        71.8        14.8
## 5 -0.7798 [-1.1319; -0.4277]         7.7        14.6
## 6 -7.0411 [-7.6671; -6.4151]         2.5        13.9
## 7 -0.0402 [-0.7118;  0.6314]         2.1        13.8
##
## Number of studies combined: k = 7
## Number of observations: o = 1365
##
##           SMD           90%-CI          z  p-value
## Common effect model -0.7403 [-0.8383; -0.6423] -12.42 < 0.0001
## Random effects model -1.6233 [-2.5648; -0.6818]  -2.84   0.0046
##
## Quantifying heterogeneity:
## tau^2 = 2.2053 [1.3566; 20.4690]; tau = 1.4850 [1.1647; 4.5243]
## I^2 = 98.0% [97.3%; 98.5%]; H = 7.03 [6.03; 8.18]
##
## Test of heterogeneity:
##           Q d.f.  p-value
```

```
## 296.18      6 < 0.0001
##
## Details on meta-analytical method:
## - Inverse variance method
## - DerSimonian-Laird estimator for tau^2
## - Jackson method for confidence interval of tau^2 and tau
## - Hedges' g (bias corrected standardised mean difference; using exact formulae)
```

```
# For the LOS outcome, all but the 7th study found a significant improvement
# with tubeless PCNL. This observation comes from the fact that none of the
# confidence intervals on the standardized mean difference include 0 except for
# the 7th study. Since this study's confidence interval includes 0, it will have
# a large p-value resulting from the null hypothesis that SMD = 0, or
# mu_trt = mu_ctrl, which means there is not significant evidence to reject said
# null. For that matter, the confidence interval was nearly centered around 0,
# further supporting this conclusion. This is additionally supported by the
# following p-values calculated for each individual study.
cat('Individual study p-values:', meta2$pval)
```

```
## Individual study p-values: 3.516396e-05 1.732071e-06 0.0418164 1.001734e-12 0.0002693254 2.021064e-7
```

```
# All studies are significant at alpha = 0.1 aside from the seventh study, with
# a p-value larger than 0.9
```

```
# b
DL_tau2 = metacont(e_n2, e_mean2, e_sd2, c_n2, c_mean2, c_sd2, sm="SMD",
                  method.tau="DL")$tau # ask for tau
PM_tau2 = metacont(e_n2, e_mean2, e_sd2, c_n2, c_mean2, c_sd2, sm="SMD",
                  method.tau="PM")$tau
ML_tau2 = metacont(e_n2, e_mean2, e_sd2, c_n2, c_mean2, c_sd2, sm="SMD",
                  method.tau="ML")$tau
lower_tau2 = c(1.1647, 1.5224, 1.5224) # low 95% estimates for tau, each method
upper_tau2 = c(4.5243, 5.339, 5.339)
tau_estimates2 = rbind(DL_tau2, PM_tau2, ML_tau2) # combine tau estimates
summary2 = data.frame(tau_estimates2, lower_tau2, upper_tau2) # combine all
summary2
```

```
##          tau_estimates2 lower_tau2 upper_tau2
## DL_tau2          1.485035      1.1647      4.5243
## PM_tau2          2.402373      1.5224      5.3390
## ML_tau2          2.209389      1.5224      5.3390
```

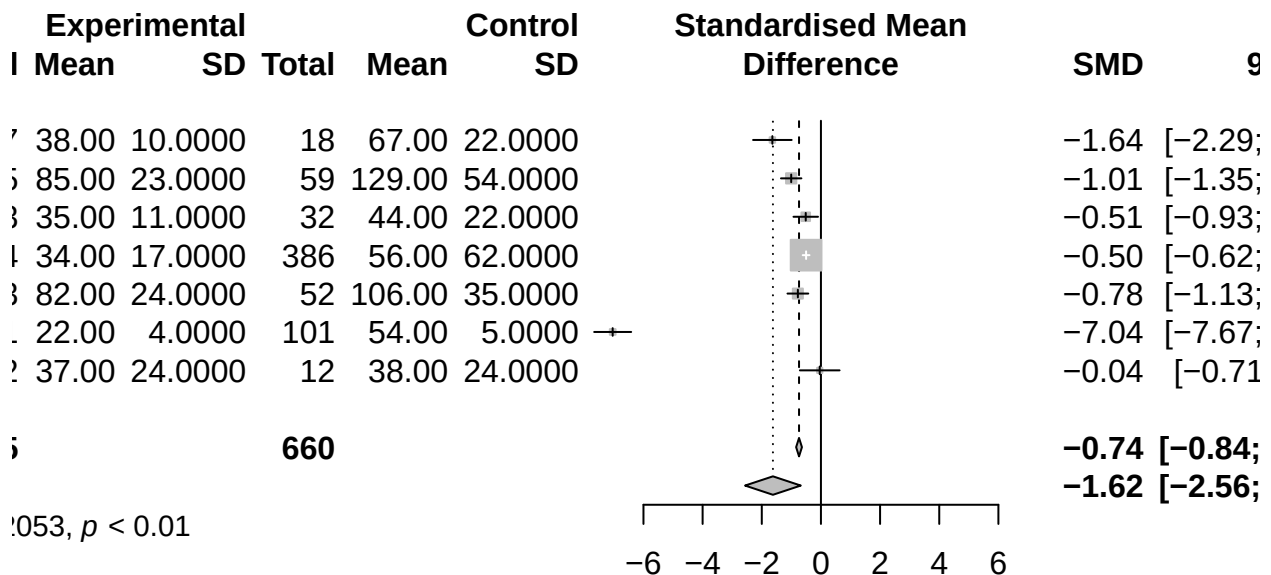
```
# There are not any major differences between the tau^2 estimates, namely
# because none of them contain 0 in the confidence interval. If 0 were included
# in the confidence interval, it would suggest there is no significant between-
# study variability. While the confidence intervals are slightly different
# between DL and PM, and DL and ML, (due to different errors for each method)
# the methods could arguably be used interchangeably because none of the
# intervals include 0, meaning the decision of the hypothesis test likely would
# not change. It should also be noted that the confidence interval for tau is
# the same again for Paule-Mandel and maximum likelihood methods.
```

```
# c
# For i = 1,2,...k studies, theta = observed standardized mean difference
# H_0: theta_1 = theta_2 = ... = theta_k vs. H_1: at least one inequality
Q2 = metacont(e_n2, e_mean2, e_sd2, c_n2, c_mean2, c_sd2, sm="SMD",
              method.tau="DL")$Q # get Q value from metacont
cat('Cochran p-value:',pchisq(Q2,6,lower.tail=FALSE)) # calculate p-value
```

```
## Cochran p-value: 5.376775e-61
```

```
# The p-value is very small, meaning we reject the null hypothesis at alpha=0.1.
# This means there is significant between-study variability, and we should use
# a random effects model, as it takes tau^2 into account in its development.
```

```
# d using logical operator "random" in metacont()
forest(metacont(e_n2, e_mean2, e_sd2, c_n2, c_mean2, c_sd2, sm="SMD",
               method.tau="DL",level=0.9,level.ma =0.9,random=1))
```



```
# e
# The results for the LOS outcome are very similar to the Duration outcome, in
# that the model shows a significant improvement in health as a result of
# tubeless PCNL. This is because confidence interval (and thus point estimate)
# for the expected effect size is less than 0, leading to a low p-value for the
# hypothesis that the two groups have the same mean. This result is
# highly effected by the large sample size in the fourth study, which gives it
```

*# a larger weight in the meta-analysis, and thus a larger "pull" on the overall
results. To summarize, there is a statistically significant result between the
groups based on the LOS outcome.*

ANALGESIC

```
# a
e_mean3 = c(39,150,103,37,82)
e_sd3 = c(35,97,116,31,24)
e_n3 = c(45,33,454,43,101)
c_mean3 = c(75,246,250,70,126)
c_sd3 = c(32,167,132,36,33)
c_n3 = c(59,32,386,52,101)
meta3 = metacont(e_n3, e_mean3, e_sd3, c_n3, c_mean3, c_sd3, sm="SMD",
                 method.tau="DL", level=0.9, level.ma =0.9)
summary(meta3)
```

```
##           SMD           90%-CI %W(common) %W(random)
## 1 -1.0722 [-1.4208; -0.7237]      8.1      16.5
## 2 -0.6973 [-1.1184; -0.2763]      5.6      13.1
## 3 -1.1882 [-1.3117; -1.0647]     64.5      32.5
## 4 -0.9675 [-1.3264; -0.6086]      7.6      16.0
## 5 -1.5192 [-1.7825; -1.2559]     14.2      21.8
##
## Number of studies combined: k = 5
## Number of observations: o = 1306
##
##           SMD           90%-CI      z  p-value
## Common effect model -1.1817 [-1.2809; -1.0825] -19.59 < 0.0001
## Random effects model -1.1414 [-1.3304; -0.9524]  -9.93 < 0.0001
##
## Quantifying heterogeneity:
## tau^2 = 0.0349 [0.0000; 0.6280]; tau = 0.1869 [0.0000; 0.7925]
## I^2 = 56.8% [0.7%; 81.2%]; H = 1.52 [1.00; 2.31]
##
## Test of heterogeneity:
##      Q d.f. p-value
## 9.27   4  0.0548
##
## Details on meta-analytical method:
## - Inverse variance method
## - DerSimonian-Laird estimator for tau^2
## - Jackson method for confidence interval of tau^2 and tau
## - Hedges' g (bias corrected standardised mean difference; using exact formulae)
```

*# For the Analgesic outcome, none of the studies contain 0 in their effect size
confidence interval, meaning all effect sizes are significantly different to 0.
This is further supported by the following p-values.*

```
cat('Individual study p-values:', meta3$pval)
```

```
## Individual study p-values: 4.195356e-07 0.006445869 2.095778e-56 9.263284e-06 2.288977e-21
```

```
# All of these p-values are less than alpha = 0.1, meaning there is a  
# significant difference between tubeless PCNL and standard PCNL. It is worth  
# noting that the 3rd and 5th study had extremely low p-values, especially the  
# third study. This is a result of the studies having larger sample sizes and/or  
# smaller variances
```

```
# b  
DL_tau3 = metacont(e_n3, e_mean3, e_sd3, c_n3, c_mean3, c_sd3, sm="SMD",  
                  method.tau="DL")$tau # ask for tau  
PM_tau3 = metacont(e_n3, e_mean3, e_sd3, c_n3, c_mean3, c_sd3, sm="SMD",  
                  method.tau="PM")$tau  
ML_tau3 = metacont(e_n3, e_mean3, e_sd3, c_n3, c_mean3, c_sd3, sm="SMD",  
                  method.tau="ML")$tau  
lower_tau3 = c(0,0,0) # low 95% estimates for tau, each method  
upper_tau3 = c(0.7925,0.8412,0.8412)  
tau_estimates3 = rbind(DL_tau3,PM_tau3,ML_tau3) # combine tau estimates  
summary3 = data.frame(tau_estimates3,lower_tau3,upper_tau3) # combine all  
summary3
```

```
##      tau_estimates3 lower_tau3 upper_tau3  
## DL_tau3      0.1869179         0      0.7925  
## PM_tau3      0.2231211         0      0.8412  
## ML_tau3      0.1440965         0      0.8412
```

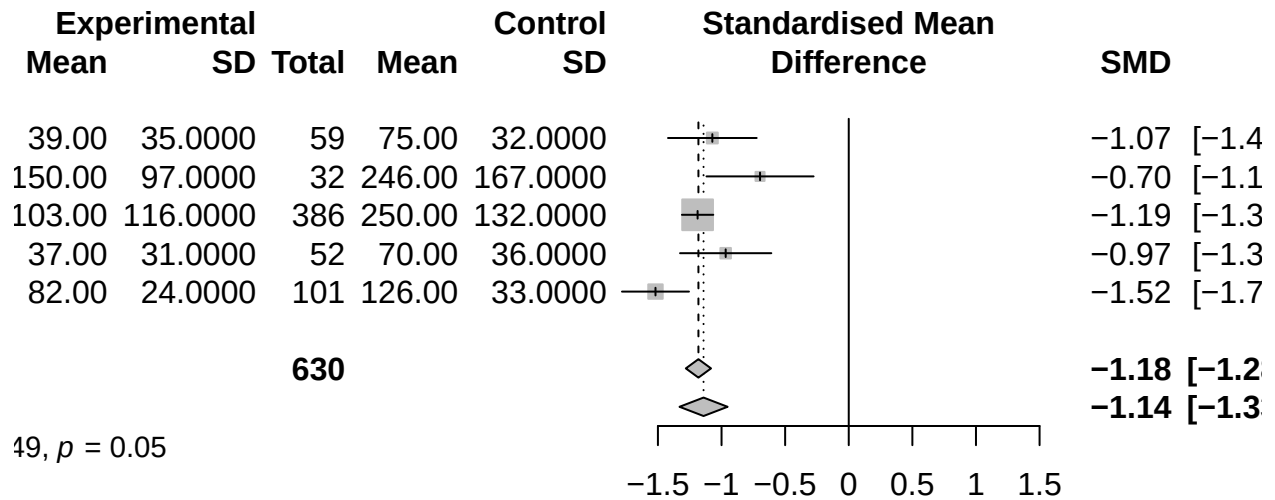
```
# The confidence intervals around tau are similar for all three methods.  
# Notably, the Paule-Mandel and maximum likelihood methods resulted in the  
# exact same confidence interval. It should be stated that all of the methods  
# included 0 in the confidence interval, which could indicate that there is not  
# significant evidence that between-study variability is greater than 0. That  
# said, there will not likely be any differences observed in the final results  
# if the methods were to be interchanged, because they are relatively close with  
# similar confidence levels. To examine this argument further, a heterogeneity  
# test should be conducted.
```

```
# c  
# For i = 1,2,..k studies, theta = observed standardized mean difference  
# H_0: theta_1 = theta_2 = ... = theta_k vs. H_1: at least one inequality  
Q3 = metacont(e_n3, e_mean3, e_sd3, c_n3, c_mean3, c_sd3, sm="SMD",  
              method.tau="DL")$Q # get Q value from metacont  
cat('Cochran p-value:',pchisq(Q3,4,lower.tail=FALSE)) # calculate p-value
```

```
## Cochran p-value: 0.05479589
```

```
# At level of significance 0.1, the null hypothesis is rejected, leading to the  
# conclusion that there is significant between-study variability. This means the  
# random effects model is more appropriate for the analysis, because the tau^2  
# is not zero and should be included .
```

```
# d using logical operator "random" in metacont()  
forest(metacont(e_n3, e_mean3, e_sd3, c_n3, c_mean3, c_sd3, sm="SMD",  
                method.tau="DL",level=0.9,level.ma =0.9,random=1))
```

e
 # Like the other outcomes, the overall conclusion is that tubeless PCNL and
 # standard PCNL result in significantly different results for the "Analgesic"
 # outcome. The conclusion is the same in that it can be said with 90% confidence
 # that tubeless PCNL improves health since the corresponding confidence interval
 # is entirely below 0.